

Provenance-Anchored Reputation: Grounding Agent Reputation in Real, Completed Payments

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Abstract

AI agents increasingly transact on behalf of their users, and an agent must often judge whether a counterparty is trustworthy before committing to an interaction. Reputation systems address this problem for human marketplaces, and several recent proposals extend reputation to agents. These proposals share a common weakness: the evidence that an interaction occurred is self-issued by the interacting agents, so a single operator that controls both sides can fabricate it. We propose to make this evidence costly to fabricate by anchoring each reputation record to an artifact that a genuine payment already produces, namely a settlement receipt signed by the buyer, the merchant, and the payment network. Under this construction, fabricating a record requires executing a real, paid transaction. We introduce Provenance-Anchored Reputation (PAR), a compact record format with an open-source reference implementation, built on the established Agent-to-Agent (A2A) and Agent Payments Protocol (AP2) standards. PAR supplies verifiable evidence of interaction; the separate task of converting that evidence into a score is delegated to dedicated scoring systems.

1 Introduction

Agents now discover one another, negotiate tasks, and settle payments through shared protocols. A2A enables agents to describe their capabilities and communicate, and AP2 enables an agent to pay using a signed authorization from its user [4, 2]. These protocols establish *who* an agent is and *that* it is authorized to pay; they say nothing about *how well* an agent has performed in the past. This omission is consequential, because a buyer’s agent must choose among competing sellers and a seller must decide which buyers to accept.

The same gap is the subject of an open design discussion in the A2A project (Discussion #1631), which proposes a shared extension for agents to publish trust signals, each referencing some underlying record of interaction [1]. The discussion leaves the nature of that record unspecified. In the proposals advanced to date, including the Agent Reputation Protocol (ARP) prototype [5], the record is self-issued by the interacting parties, either as a row appended to a local ledger or as an attestation jointly signed by both agents. Such records are forgeable by a single operator that controls both the buyer and the seller, which can synthesize an arbitrary interaction history at negligible cost. As participants in the discussion observe, resisting this attack requires evidence that an agent cannot unilaterally assert about itself.

2 Approach: reputation evidence as a costly signal

Our approach rests on a well-established principle: a signal is credible when it is costly to fake. The same principle explains why an educational credential signals ability in labor markets [7], why a costly biological display signals fitness [8], and why an online review marked “verified purchase” carries more weight than an anonymous one [6], since each is backed by a real, paid transaction.

A reputation record should therefore be expensive to produce. The least costly source of such a record is one that a genuine payment already generates. When two parties settle a payment on a modern rail, the resulting receipt is co-signed by parties that do not trust one another (the user, the merchant, and the payment network) and cannot be produced without an actual transfer of funds. Rather than defining a new proof of interaction, we anchor each reputation record to this existing, multi-party-signed receipt. Because fabricating a record then requires a real, paid transaction, the cost of manufacturing fraudulent reputation approaches the cost of earning legitimate reputation.

3 The Verified Interaction Record

The unit of PAR is a *Verified Interaction Record* (VIR). A VIR asserts that two agents, *A* and *B*, interacted in defined roles, with *A* as the service provider and *B* as the buyer, and a stated outcome, and it references the signed payment receipt for that transaction. Issuing a VIR requires possession of such a receipt, which entails that a real payment occurred.

A VIR comprises four components.

- **Anchor.** A reference to the signed payment receipt together with a verifiable fingerprint (a content hash) of it. The receipt must separate *what was purchased* from *how it was paid*. AP2’s Checkout Mandate is the first such receipt; a card-network receipt or a bank-transfer confirmation would serve equivalently.
- **Parties.** The two agents, each identified by a durable identity (for example, a decentralized identifier) rather than a short-lived process. Reputation therefore accrues to the operator behind the agent: a negative record cannot be discarded by creating a new agent, and it is never bound to the human customer.
- **Interaction class.** A short label for the task (for example, “flight booking”), so that a record concerning one capability is not misread as evidence about another.
- **Outcome.** Whether the payment settled, and whether it was subsequently reversed.

A VIR carries only item-level facts (what was purchased, and from whom); it never contains payment-instrument data or the customer’s personal information. This separation follows directly from the receipt’s own split between the purchase and the payment.

Outcomes are interpreted with a single distinction. A payment reversed through a dispute (a chargeback) indicates that the interaction did not succeed, and the corresponding record is invalidated. A refund is treated differently: the interaction took place and was undone by mutual agreement, which is frequently a sign of good conduct by the seller, so the record remains valid evidence.

4 Reference implementation

We provide an open-source reference implementation comprising the record format, the validation rules that enforce its invariants, and the procedure that derives a VIR from a genuine AP2 receipt.

The validators enforce that a record references a real, signed receipt; that it names durable operator identities; that it contains no payment or personal data; that the two agents occupy opposite roles; and that a reversed payment invalidates the record. A bounded formal model in TLA⁺ mechanically checks the most important of these invariants. The implementation, the model, and the A2A extension format are publicly available.¹

5 Position in the stack and related work

PAR occupies a single layer within a larger stack (Figure 1). Beneath it, the core standards (A2A and AP2) supply identity and signed payments. PAR sits directly above them and converts a settled payment into an item of reputation evidence. Above PAR, independent scoring systems determine how to aggregate many records into a score: how to weight recent behavior, how to detect adversarial activity, and how to combine signals. These design choices are varied and contested, and we deliberately do not commit to any one of them; PAR’s role is confined to making the evidence beneath those scores trustworthy.

This also defines PAR’s relationship to the A2A trust discussion: it is a concrete answer to that discussion’s open question of what a trust signal should reference. The closest existing construction is ERC-8004, which likewise restricts feedback to paying customers but records it on a blockchain and on a bare payment proof [3]. PAR requires no blockchain and preserves the full purchase context behind each record.

¹Reference implementation: <https://github.com/ravikiran438/par-reputation> (Apache-2.0).

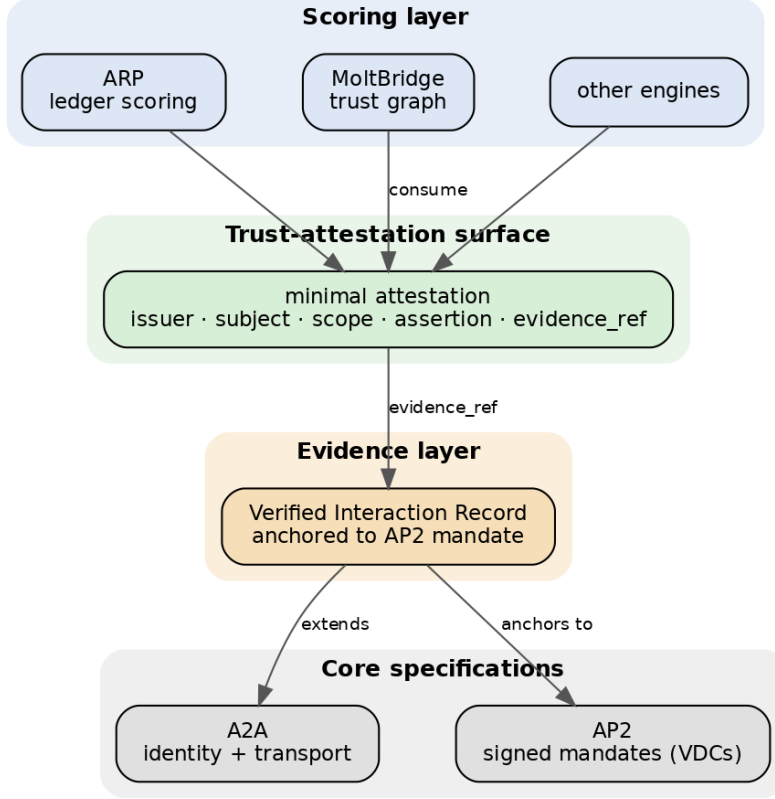


Figure 1: Where PAR sits. The core standards provide identity and signed payments; PAR turns a settled payment into a reputation record; separate scoring systems sit on top.

6 Scope and limitations

PAR raises the cost of fabricating reputation; it does not render fabrication impossible. An adversary willing to execute real, paid transactions can still accumulate standing, which is the intended outcome, since at that point the adversary incurs roughly the cost borne by an honest seller. Detecting collusion distributed across many accounts, and converting records into scores, are responsibilities of the scoring layer above PAR rather than of the evidence layer itself. Finally, PAR depends on a payment rail that issues receipts with the required structure; AP2 is the first such rail, and we anticipate others.

7 Conclusion

The contribution is deliberately narrow. Rather than defining a new proof that two agents interacted, PAR reuses the signed receipt that a genuine payment already produces, rendering a reputation record costly to fabricate. The result is a thin, off-chain evidence layer for agent reputation, built entirely on existing standards and accompanied by a working reference implementation. The harder and more plural problem of scoring is left to the systems designed for it.

References

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